

Practical 8:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n", (float)hit/n*100, (float)fault/n*100);

    return 0;
}
```

Output:



```
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ nano practical8_lru.c  
  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ gcc practical8_lru.c -o lru  
./lru  
Page    Frames  
1        1 - -  
2        1 2 -  
3        1 2 3  
4        4 2 3  
1        4 1 3  
2        4 1 2  
5        5 1 2  
1        5 1 2  
2        5 1 2  
3        3 1 2  
4        3 4 2  
5        3 4 5  
  
Total Hits = 2  
Total Faults = 10  
Hit Percentage = 16.67%  
Fault Percentage = 83.33%  
  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$
```